

```
In [22]: import pandas as pd
import numpy as np

url = "https://raw.githubusercontent.com/SR1608/Datasets/main/covid-data."
var = pd.read_csv(url)
var.head()
```

```
Out[22]:
```

|   | iso_code | continent | location    | date     | total_cases | new_cases | new_cases_ |
|---|----------|-----------|-------------|----------|-------------|-----------|------------|
| 0 | AFG      | Asia      | Afghanistan | 31/12/19 | NaN         | 0.0       |            |
| 1 | AFG      | Asia      | Afghanistan | 01/01/20 | NaN         | 0.0       |            |
| 2 | AFG      | Asia      | Afghanistan | 02/01/20 | NaN         | 0.0       |            |
| 3 | AFG      | Asia      | Afghanistan | 03/01/20 | NaN         | 0.0       |            |
| 4 | AFG      | Asia      | Afghanistan | 04/01/20 | NaN         | 0.0       |            |

```
In [ ]: #Load data
```

```
In [12]: pd.set_option('display.max_rows',None)
pd.set_option('display.max_column',None)
print(var.iloc[:])
```

|    | mpg  | cylinders | displacement | horsepower | weight | acceleration | year |
|----|------|-----------|--------------|------------|--------|--------------|------|
| 0  | 18.0 | 8         | 307.0        | 130        | 3504   | 12.0         | 70   |
| 1  | 15.0 | 8         | 350.0        | 165        | 3693   | 11.5         | 70   |
| 2  | 18.0 | 8         | 318.0        | 150        | 3436   | 11.0         | 70   |
| 3  | 16.0 | 8         | 304.0        | 150        | 3433   | 12.0         | 70   |
| 4  | 17.0 | 8         | 302.0        | 140        | 3449   | 10.5         | 70   |
| 5  | 15.0 | 8         | 429.0        | 198        | 4341   | 10.0         | 70   |
| 6  | 14.0 | 8         | 454.0        | 220        | 4354   | 9.0          | 70   |
| 7  | 14.0 | 8         | 440.0        | 215        | 4312   | 8.5          | 70   |
| 8  | 14.0 | 8         | 455.0        | 225        | 4425   | 10.0         | 70   |
| 9  | 15.0 | 8         | 390.0        | 190        | 3850   | 8.5          | 70   |
| 10 | 15.0 | 8         | 383.0        | 170        | 3563   | 10.0         | 70   |
| 11 | 14.0 | 8         | 340.0        | 160        | 3609   | 8.0          | 70   |
| 12 | 15.0 | 8         | 400.0        | 150        | 3761   | 9.5          | 70   |
| 13 | 14.0 | 8         | 455.0        | 225        | 3086   | 10.0         | 70   |
| 14 | 24.0 | 4         | 113.0        | 95         | 2372   | 15.0         | 70   |
| 15 | 22.0 | 6         | 198.0        | 95         | 2833   | 15.5         | 70   |
| 16 | 18.0 | 6         | 199.0        | 97         | 2774   | 15.5         | 70   |
| 17 | 21.0 | 6         | 200.0        | 85         | 2587   | 16.0         | 70   |
| 18 | 27.0 | 4         | 97.0         | 88         | 2130   | 14.5         | 70   |
| 19 | 26.0 | 4         | 97.0         | 46         | 1835   | 20.5         | 70   |
| 20 | 25.0 | 4         | 110.0        | 87         | 2672   | 17.5         | 70   |
| 21 | 24.0 | 4         | 107.0        | 90         | 2430   | 14.5         | 70   |
| 22 | 25.0 | 4         | 104.0        | 95         | 2375   | 17.5         | 70   |
| 23 | 26.0 | 4         | 121.0        | 113        | 2234   | 12.5         | 70   |

|    |      |   |       |     |      |      |    |
|----|------|---|-------|-----|------|------|----|
| 27 | 11.0 | 8 | 318.0 | 210 | 4382 | 13.5 | 70 |
| 28 | 9.0  | 8 | 304.0 | 193 | 4732 | 18.5 | 70 |
| 29 | 27.0 | 4 | 97.0  | 88  | 2130 | 14.5 | 71 |
| 30 | 28.0 | 4 | 140.0 | 90  | 2264 | 15.5 | 71 |
| 31 | 25.0 | 4 | 113.0 | 95  | 2228 | 14.0 | 71 |
| 32 | 19.0 | 6 | 232.0 | 100 | 2634 | 13.0 | 71 |
| 33 | 16.0 | 6 | 225.0 | 105 | 3439 | 15.5 | 71 |
| 34 | 17.0 | 6 | 250.0 | 100 | 3329 | 15.5 | 71 |
| 35 | 19.0 | 6 | 250.0 | 88  | 3302 | 15.5 | 71 |
| 36 | 18.0 | 6 | 232.0 | 100 | 3288 | 15.5 | 71 |
| 37 | 14.0 | 8 | 350.0 | 165 | 4209 | 12.0 | 71 |
| 38 | 14.0 | 8 | 400.0 | 175 | 4464 | 11.5 | 71 |
| 39 | 14.0 | 8 | 351.0 | 153 | 4154 | 13.5 | 71 |
| 40 | 14.0 | 8 | 318.0 | 150 | 4096 | 13.0 | 71 |
| 41 | 12.0 | 8 | 383.0 | 180 | 4955 | 11.5 | 71 |
| 42 | 13.0 | 8 | 400.0 | 170 | 4746 | 12.0 | 71 |
| 43 | 13.0 | 8 | 400.0 | 175 | 5140 | 12.0 | 71 |
| 44 | 18.0 | 6 | 258.0 | 110 | 2962 | 13.5 | 71 |
| 45 | 22.0 | 4 | 140.0 | 72  | 2408 | 19.0 | 71 |
| 46 | 19.0 | 6 | 250.0 | 100 | 3282 | 15.0 | 71 |
| 47 | 18.0 | 6 | 250.0 | 88  | 3139 | 14.5 | 71 |
| 48 | 23.0 | 4 | 122.0 | 86  | 2220 | 14.0 | 71 |
| 49 | 28.0 | 4 | 116.0 | 90  | 2123 | 14.0 | 71 |
| 50 | 30.0 | 4 | 79.0  | 70  | 2074 | 19.5 | 71 |
| 51 | 30.0 | 4 | 88.0  | 76  | 2065 | 14.5 | 71 |
| 52 | 31.0 | 4 | 71.0  | 65  | 1773 | 19.0 | 71 |
| 53 | 35.0 | 4 | 72.0  | 69  | 1613 | 18.0 | 71 |
| 54 | 27.0 | 4 | 97.0  | 60  | 1834 | 19.0 | 71 |
| 55 | 26.0 | 4 | 91.0  | 70  | 1955 | 20.5 | 71 |
| 56 | 24.0 | 4 | 113.0 | 95  | 2278 | 15.5 | 72 |
| 57 | 25.0 | 4 | 97.5  | 80  | 2126 | 17.0 | 72 |
| 58 | 23.0 | 4 | 97.0  | 54  | 2254 | 23.5 | 72 |
| 59 | 20.0 | 4 | 140.0 | 90  | 2408 | 19.5 | 72 |
| 60 | 21.0 | 4 | 122.0 | 86  | 2226 | 16.5 | 72 |
| 61 | 13.0 | 8 | 350.0 | 165 | 4274 | 12.0 | 72 |
| 62 | 14.0 | 8 | 400.0 | 175 | 4385 | 12.0 | 72 |
| 63 | 15.0 | 8 | 318.0 | 150 | 4135 | 13.5 | 72 |
| 64 | 14.0 | 8 | 351.0 | 153 | 4129 | 13.0 | 72 |
| 65 | 17.0 | 8 | 304.0 | 150 | 3672 | 11.5 | 72 |
| 66 | 11.0 | 8 | 429.0 | 208 | 4633 | 11.0 | 72 |
| 67 | 13.0 | 8 | 350.0 | 155 | 4502 | 13.5 | 72 |
| 68 | 12.0 | 8 | 350.0 | 160 | 4456 | 13.5 | 72 |
| 69 | 13.0 | 8 | 400.0 | 190 | 4422 | 12.5 | 72 |
| 70 | 19.0 | 3 | 70.0  | 97  | 2330 | 13.5 | 72 |
| 71 | 15.0 | 8 | 304.0 | 150 | 3892 | 12.5 | 72 |
| 72 | 13.0 | 8 | 307.0 | 130 | 4098 | 14.0 | 72 |
| 73 | 13.0 | 8 | 302.0 | 140 | 4294 | 16.0 | 72 |
| 74 | 14.0 | 8 | 318.0 | 150 | 4077 | 14.0 | 72 |
| 75 | 18.0 | 4 | 121.0 | 112 | 2933 | 14.5 | 72 |
| 76 | 22.0 | 4 | 121.0 | 76  | 2511 | 18.0 | 72 |

|     |   |                                   |
|-----|---|-----------------------------------|
| 375 | 3 | honda accord                      |
| 376 | 3 | toyota corolla                    |
| 377 | 3 | honda civic                       |
| 378 | 3 | honda civic (auto)                |
| 379 | 3 | datson 310 gx                     |
| 380 | 1 | buick century limited             |
| 381 | 1 | oldsmobile cutlass ciera (diesel) |
| 382 | 1 | chrysler lebaron medallion        |
| 383 | 1 | ford granada l                    |
| 384 | 3 | toyota celica gt                  |
| 385 | 1 | dodge charger 2.2                 |
| 386 | 1 | chevrolet c aro                   |
| 387 | 1 | ford mustang gl                   |
| 388 | 2 | vw pickup                         |
| 389 | 1 | dodge r page                      |
| 390 | 1 | ford ranger                       |
| 391 | 1 | chevy s-10                        |

```
In [ ]: # High Level Data Understanding:
```

```
In [ ]: #a. Find no. of rows & columns in the dataset
```

```
In [24]: var.shape
```

```
Out[24]: (57394, 49)
```

```
In [ ]: #b. Data types of columns.
```

```
In [26]: var.dtypes
```

|                                    |         |
|------------------------------------|---------|
| location                           | object  |
| date                               | object  |
| total_cases                        | float64 |
| new_cases                          | float64 |
| new_cases_smoothed                 | float64 |
| total_deaths                       | float64 |
| new_deaths                         | float64 |
| new_deaths_smoothed                | float64 |
| total_cases_per_million            | float64 |
| new_cases_per_million              | float64 |
| new_cases_smoothed_per_million     | float64 |
| total_deaths_per_million           | float64 |
| new_deaths_per_million             | float64 |
| new_deaths_smoothed_per_million    | float64 |
| reproduction_rate                  | float64 |
| icu_patients                       | float64 |
| icu_patients_per_million           | float64 |
| hosp_patients                      | float64 |
| hosp_patients_per_million          | float64 |
| weekly_icu_admissions              | float64 |
| weekly_icu_admissions_per_million  | float64 |
| weekly_hosp_admissions             | float64 |
| weekly_hosp_admissions_per_million | float64 |
| total_tests                        | float64 |
| new_tests                          | float64 |
| total_tests_per_thousand           | float64 |
| new_tests_per_thousand             | float64 |
| new_tests_smoothed                 | float64 |
| new_tests_smoothed_per_thousand    | float64 |
| tests_per_case                     | float64 |
| positive_rate                      | float64 |
| stringency_index                   | float64 |
| population                         | float64 |
| population_density                 | float64 |
| median_age                         | float64 |
| aged_65_older                      | float64 |
| aged_70_older                      | float64 |
| gdp_per_capita                     | float64 |
| extreme_poverty                    | float64 |
| cardiovasc_death_rate              | float64 |
| diabetes_prevalence                | float64 |
| female_smokers                     | float64 |
| male_smokers                       | float64 |
| handwashing_facilities             | float64 |
| hospital_beds_per_thousand         | float64 |
| life_expectancy                    | float64 |
| human_develo ent_index             | float64 |
| dtype:                             | object  |

```
In [ ]: #c. Info & describe of data in datafr e.
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 57394 entries, 0 to 57393
Data columns (total 49 columns):
```

| #  | Column                             | Non-Null Count | Dtype   |
|----|------------------------------------|----------------|---------|
| 0  | iso_code                           | 57071 non-null | object  |
| 1  | continent                          | 56748 non-null | object  |
| 2  | location                           | 57394 non-null | object  |
| 3  | date                               | 57394 non-null | object  |
| 4  | total_cases                        | 53758 non-null | float64 |
| 5  | new_cases                          | 56465 non-null | float64 |
| 6  | new_cases_smoothed                 | 55652 non-null | float64 |
| 7  | total_deaths                       | 44368 non-null | float64 |
| 8  | new_deaths                         | 56465 non-null | float64 |
| 9  | new_deaths_smoothed                | 55652 non-null | float64 |
| 10 | total_cases_per_million            | 53471 non-null | float64 |
| 11 | new_cases_per_million              | 56401 non-null | float64 |
| 12 | new_cases_smoothed_per_million     | 55587 non-null | float64 |
| 13 | total_deaths_per_million           | 44096 non-null | float64 |
| 14 | new_deaths_per_million             | 56401 non-null | float64 |
| 15 | new_deaths_smoothed_per_million    | 55587 non-null | float64 |
| 16 | reproduction_rate                  | 37696 non-null | float64 |
| 17 | icu_patients                       | 4490 non-null  | float64 |
| 18 | icu_patients_per_million           | 4490 non-null  | float64 |
| 19 | hosp_patients                      | 5005 non-null  | float64 |
| 20 | hosp_patients_per_million          | 5005 non-null  | float64 |
| 21 | weekly_icu_admissions              | 357 non-null   | float64 |
| 22 | weekly_icu_admissions_per_million  | 357 non-null   | float64 |
| 23 | weekly_hosp_admissions             | 645 non-null   | float64 |
| 24 | weekly_hosp_admissions_per_million | 645 non-null   | float64 |
| 25 | total_tests                        | 22017 non-null | float64 |
| 26 | new_tests                          | 21787 non-null | float64 |
| 27 | total_tests_per_thousand           | 22017 non-null | float64 |
| 28 | new_tests_per_thousand             | 21787 non-null | float64 |
| 29 | new_tests_smoothed                 | 24612 non-null | float64 |
| 30 | new_tests_smoothed_per_thousand    | 24612 non-null | float64 |
| 31 | tests_per_case                     | 22802 non-null | float64 |
| 32 | positive_rate                      | 23211 non-null | float64 |
| 33 | stringency_index                   | 47847 non-null | float64 |
| 34 | population                         | 57071 non-null | float64 |
| 35 | population_density                 | 54371 non-null | float64 |
| 36 | median_age                         | 51034 non-null | float64 |
| 37 | aged_65_older                      | 50265 non-null | float64 |
| 38 | aged_70_older                      | 50768 non-null | float64 |
| 39 | gdp_per_capita                     | 50367 non-null | float64 |
| 40 | extreme_poverty                    | 33571 non-null | float64 |
| 41 | cardiovasc_death_rate              | 51013 non-null | float64 |
| 42 | diabetes_prevalence                | 52881 non-null | float64 |
| 43 | female_smokers                     | 39669 non-null | float64 |
| 44 | male_smokers                       | 39156 non-null | float64 |

dtypes: float64(45), object(4)  
memory usage: 21.5+ MB

```
In [30]: var.describe()
```

```
Out[30]:
```

|       | total_cases  | new_cases     | new_cases_smoothed | total_deaths | new_   |
|-------|--------------|---------------|--------------------|--------------|--------|
| count | 5.375800e+04 | 56465.000000  | 55652.000000       | 4.436800e+04 | 56465. |
| mean  | 1.677974e+05 | 1953.576941   | 1920.431953        | 6.858639e+03 | 47     |
| std   | 1.693038e+06 | 18269.650340  | 17777.391785       | 5.578081e+04 | 390.   |
| min   | 1.000000e+00 | -8261.000000  | -552.000000        | 1.000000e+00 | -1918. |
| 25%   | 1.800000e+02 | 0.000000      | 0.857000           | 1.300000e+01 | 0.     |
| 50%   | 2.070000e+03 | 14.000000     | 19.429000          | 8.400000e+01 | 0.     |
| 75%   | 2.235675e+04 | 235.000000    | 245.286000         | 7.270000e+02 | 4.     |
| max   | 5.515465e+07 | 646281.000000 | 584981.857000      | 1.328537e+06 | 10600. |

```
In [ ]: #3. Low Level Data Understanding :  
  
#a. Find count of unique values in location column.
```

```
In [32]: var.location.nunique()
```

```
Out[32]: 216
```

```
In [ ]: #b. Find which continent has maximum frequency using values
```

```
In [49]: var['continent'].value_counts().idxmax()
```

```
Out[49]: 'Europe'
```

```
In [51]: var['continent'].value_counts().max()
```

```
Out[51]: 14828
```

```
In [ ]: #c. Find maximum & mean value in 'total_cases'.
```

```
In [65]: max1 = var['total_cases'].max()  
min1=var['total_cases'].mean()  
print("maximum values in total_case column is :",max1)  
  
print("mean value in total_case column is :",min1)
```

maximum values in total\_case column is : 55154651.0

```
In [70]: var['total_deaths'].describe()
```

```
Out[70]: count      4.436800e+04
mean        6.858639e+03
std         5.578081e+04
min         1.000000e+00
25%         1.300000e+01
50%         8.400000e+01
75%         7.270000e+02
max         1.328537e+06
N e: total_deaths, dtype: float64
```

```
In [ ]: # e.Find which continent has maximum 'human_development_index'.
```

```
In [88]: num1 = num1 = var.loc[var['human_development_index'].idxmax(), ['continent', 'human_development_index']]
print(num1)
```

```
continent      Europe
human_development_index  0.953
N e: 38632, dtype: object
```

```
In [90]: #4. Filter the dataframe with only these columns
#['continent', 'location', 'date', 'total_cases', 'total_deaths', 'gdp_per_capita', 'human_development_index'] and update the dataframe.
```

```
In [129]: var = var[['continent', 'location', 'date', 'total_cases', 'total_deaths', 'gdp_per_capita', 'human_development_index']]
var.head()
```

```
Out[129]:
```

|   | continent | location    | date     | total_cases | total_deaths | gdp_per_capita | human_development_index |
|---|-----------|-------------|----------|-------------|--------------|----------------|-------------------------|
| 0 | Asia      | Afghanistan | 31/12/19 | NaN         | NaN          | 1803.987       | 0.501                   |
| 1 | Asia      | Afghanistan | 01/01/20 | NaN         | NaN          | 1803.987       | 0.501                   |
| 2 | Asia      | Afghanistan | 02/01/20 | NaN         | NaN          | 1803.987       | 0.501                   |
| 3 | Asia      | Afghanistan | 03/01/20 | NaN         | NaN          | 1803.987       | 0.501                   |
| 4 | Asia      | Afghanistan | 04/01/20 | NaN         | NaN          | 1803.987       | 0.501                   |

```
In [ ]: #5. Data Cleaning
```

```
In [ ]: #a. Remove all duplicate observations
```

```
In [158]: var.drop_duplicates(inplace=True)
```

|   |      |             |          |     |     |          |
|---|------|-------------|----------|-----|-----|----------|
| 0 | Asia | Afghanistan | 31/12/19 | NaN | NaN | 1803.987 |
| 1 | Asia | Afghanistan | 01/01/20 | NaN | NaN | 1803.987 |
| 2 | Asia | Afghanistan | 02/01/20 | NaN | NaN | 1803.987 |
| 3 | Asia | Afghanistan | 03/01/20 | NaN | NaN | 1803.987 |
| 4 | Asia | Afghanistan | 04/01/20 | NaN | NaN | 1803.987 |

In [159... `var.head()`

Out[159... 

|   | continent | location    | date     | total_cases | total_deaths | gdp_per_capita | hun |
|---|-----------|-------------|----------|-------------|--------------|----------------|-----|
| 0 | Asia      | Afghanistan | 31/12/19 | NaN         | NaN          | 1803.987       |     |
| 1 | Asia      | Afghanistan | 01/01/20 | NaN         | NaN          | 1803.987       |     |
| 2 | Asia      | Afghanistan | 02/01/20 | NaN         | NaN          | 1803.987       |     |
| 3 | Asia      | Afghanistan | 03/01/20 | NaN         | NaN          | 1803.987       |     |
| 4 | Asia      | Afghanistan | 04/01/20 | NaN         | NaN          | 1803.987       |     |

In [ ]: `#Find missing values in all columns`

In [165... `count1=var.isnull().sum()`

In [167... `print(count1)`

```
continent          646
location           0
date               0
total_cases        3636
total_deaths       13026
gdp_per_capita     7027
human_develo ent_index 8147
dtype: int64
```

In [196... `var.dropna(subset=['continent'], inplace=True)`  
`var.head(100) #c. Remove all observations where continent column value is missing`

Out[196... 

|   | continent | location    | date     | total_cases | total_deaths | gdp_per_capita | hu |
|---|-----------|-------------|----------|-------------|--------------|----------------|----|
| 0 | Asia      | Afghanistan | 31/12/19 | NaN         | NaN          | 1803.987       |    |
| 1 | Asia      | Afghanistan | 01/01/20 | NaN         | NaN          | 1803.987       |    |
| 2 | Asia      | Afghanistan | 02/01/20 | NaN         | NaN          | 1803.987       |    |

|    |      |             |          |     |     |          |
|----|------|-------------|----------|-----|-----|----------|
| 5  | Asia | Afghanistan | 05/01/20 | NaN | NaN | 1803.987 |
| 6  | Asia | Afghanistan | 06/01/20 | NaN | NaN | 1803.987 |
| 7  | Asia | Afghanistan | 07/01/20 | NaN | NaN | 1803.987 |
| 8  | Asia | Afghanistan | 08/01/20 | NaN | NaN | 1803.987 |
| 9  | Asia | Afghanistan | 09/01/20 | NaN | NaN | 1803.987 |
| 10 | Asia | Afghanistan | 10/01/20 | NaN | NaN | 1803.987 |
| 11 | Asia | Afghanistan | 11/01/20 | NaN | NaN | 1803.987 |
| 12 | Asia | Afghanistan | 12/01/20 | NaN | NaN | 1803.987 |
| 13 | Asia | Afghanistan | 13/01/20 | NaN | NaN | 1803.987 |
| 14 | Asia | Afghanistan | 14/01/20 | NaN | NaN | 1803.987 |
| 15 | Asia | Afghanistan | 15/01/20 | NaN | NaN | 1803.987 |
| 16 | Asia | Afghanistan | 16/01/20 | NaN | NaN | 1803.987 |
| 17 | Asia | Afghanistan | 17/01/20 | NaN | NaN | 1803.987 |
| 18 | Asia | Afghanistan | 18/01/20 | NaN | NaN | 1803.987 |
| 19 | Asia | Afghanistan | 19/01/20 | NaN | NaN | 1803.987 |
| 20 | Asia | Afghanistan | 20/01/20 | NaN | NaN | 1803.987 |
| 21 | Asia | Afghanistan | 21/01/20 | NaN | NaN | 1803.987 |
| 22 | Asia | Afghanistan | 22/01/20 | NaN | NaN | 1803.987 |
| 23 | Asia | Afghanistan | 23/01/20 | NaN | NaN | 1803.987 |
| 24 | Asia | Afghanistan | 24/01/20 | NaN | NaN | 1803.987 |
| 25 | Asia | Afghanistan | 25/01/20 | NaN | NaN | 1803.987 |
| 26 | Asia | Afghanistan | 26/01/20 | NaN | NaN | 1803.987 |
| 27 | Asia | Afghanistan | 27/01/20 | NaN | NaN | 1803.987 |
| 28 | Asia | Afghanistan | 28/01/20 | NaN | NaN | 1803.987 |
| 29 | Asia | Afghanistan | 29/01/20 | NaN | NaN | 1803.987 |
| 30 | Asia | Afghanistan | 30/01/20 | NaN | NaN | 1803.987 |
| 31 | Asia | Afghanistan | 31/01/20 | NaN | NaN | 1803.987 |
| 32 | Asia | Afghanistan | 01/02/20 | NaN | NaN | 1803.987 |
| 33 | Asia | Afghanistan | 02/02/20 | NaN | NaN | 1803.987 |
| 34 | Asia | Afghanistan | 03/02/20 | NaN | NaN | 1803.987 |

|    |      |             |          |     |     |          |
|----|------|-------------|----------|-----|-----|----------|
| 37 | Asia | Afghanistan | 06/02/20 | NaN | NaN | 1803.987 |
| 38 | Asia | Afghanistan | 07/02/20 | NaN | NaN | 1803.987 |
| 39 | Asia | Afghanistan | 08/02/20 | NaN | NaN | 1803.987 |
| 40 | Asia | Afghanistan | 09/02/20 | NaN | NaN | 1803.987 |
| 41 | Asia | Afghanistan | 10/02/20 | NaN | NaN | 1803.987 |
| 42 | Asia | Afghanistan | 11/02/20 | NaN | NaN | 1803.987 |
| 43 | Asia | Afghanistan | 12/02/20 | NaN | NaN | 1803.987 |
| 44 | Asia | Afghanistan | 13/02/20 | NaN | NaN | 1803.987 |
| 45 | Asia | Afghanistan | 14/02/20 | NaN | NaN | 1803.987 |
| 46 | Asia | Afghanistan | 15/02/20 | NaN | NaN | 1803.987 |
| 47 | Asia | Afghanistan | 16/02/20 | NaN | NaN | 1803.987 |
| 48 | Asia | Afghanistan | 17/02/20 | NaN | NaN | 1803.987 |
| 49 | Asia | Afghanistan | 18/02/20 | NaN | NaN | 1803.987 |
| 50 | Asia | Afghanistan | 19/02/20 | NaN | NaN | 1803.987 |
| 51 | Asia | Afghanistan | 20/02/20 | NaN | NaN | 1803.987 |
| 52 | Asia | Afghanistan | 21/02/20 | NaN | NaN | 1803.987 |
| 53 | Asia | Afghanistan | 22/02/20 | NaN | NaN | 1803.987 |
| 54 | Asia | Afghanistan | 23/02/20 | NaN | NaN | 1803.987 |
| 55 | Asia | Afghanistan | 24/02/20 | NaN | NaN | 1803.987 |
| 56 | Asia | Afghanistan | 25/02/20 | 1.0 | NaN | 1803.987 |
| 57 | Asia | Afghanistan | 26/02/20 | 1.0 | NaN | 1803.987 |
| 58 | Asia | Afghanistan | 27/02/20 | 1.0 | NaN | 1803.987 |
| 59 | Asia | Afghanistan | 28/02/20 | 1.0 | NaN | 1803.987 |
| 60 | Asia | Afghanistan | 29/02/20 | 1.0 | NaN | 1803.987 |
| 61 | Asia | Afghanistan | 01/03/20 | 1.0 | NaN | 1803.987 |
| 62 | Asia | Afghanistan | 02/03/20 | 1.0 | NaN | 1803.987 |
| 63 | Asia | Afghanistan | 03/03/20 | NaN | NaN | 1803.987 |
| 64 | Asia | Afghanistan | 04/03/20 | NaN | NaN | 1803.987 |
| 65 | Asia | Afghanistan | 05/03/20 | NaN | NaN | 1803.987 |

|    |      |             |          |       |     |          |
|----|------|-------------|----------|-------|-----|----------|
| 68 | Asia | Afghanistan | 08/03/20 | 4.0   | NaN | 1803.987 |
| 69 | Asia | Afghanistan | 09/03/20 | NaN   | NaN | 1803.987 |
| 70 | Asia | Afghanistan | 10/03/20 | NaN   | NaN | 1803.987 |
| 71 | Asia | Afghanistan | 11/03/20 | 7.0   | NaN | 1803.987 |
| 72 | Asia | Afghanistan | 12/03/20 | NaN   | NaN | 1803.987 |
| 73 | Asia | Afghanistan | 13/03/20 | NaN   | NaN | 1803.987 |
| 74 | Asia | Afghanistan | 14/03/20 | NaN   | NaN | 1803.987 |
| 75 | Asia | Afghanistan | 15/03/20 | 10.0  | NaN | 1803.987 |
| 76 | Asia | Afghanistan | 16/03/20 | 16.0  | NaN | 1803.987 |
| 77 | Asia | Afghanistan | 17/03/20 | 21.0  | NaN | 1803.987 |
| 78 | Asia | Afghanistan | 18/03/20 | 22.0  | NaN | 1803.987 |
| 79 | Asia | Afghanistan | 19/03/20 | 22.0  | NaN | 1803.987 |
| 80 | Asia | Afghanistan | 20/03/20 | 22.0  | NaN | 1803.987 |
| 81 | Asia | Afghanistan | 21/03/20 | 24.0  | NaN | 1803.987 |
| 82 | Asia | Afghanistan | 22/03/20 | 24.0  | NaN | 1803.987 |
| 83 | Asia | Afghanistan | 23/03/20 | 34.0  | NaN | 1803.987 |
| 84 | Asia | Afghanistan | 24/03/20 | 40.0  | 1.0 | 1803.987 |
| 85 | Asia | Afghanistan | 25/03/20 | 42.0  | 1.0 | 1803.987 |
| 86 | Asia | Afghanistan | 26/03/20 | 75.0  | 1.0 | 1803.987 |
| 87 | Asia | Afghanistan | 27/03/20 | 75.0  | 1.0 | 1803.987 |
| 88 | Asia | Afghanistan | 28/03/20 | 91.0  | 2.0 | 1803.987 |
| 89 | Asia | Afghanistan | 29/03/20 | 106.0 | 3.0 | 1803.987 |
| 90 | Asia | Afghanistan | 30/03/20 | 114.0 | 4.0 | 1803.987 |
| 91 | Asia | Afghanistan | 31/03/20 | 141.0 | 4.0 | 1803.987 |
| 92 | Asia | Afghanistan | 01/04/20 | 166.0 | 4.0 | 1803.987 |
| 93 | Asia | Afghanistan | 02/04/20 | 192.0 | 4.0 | 1803.987 |
| 94 | Asia | Afghanistan | 03/04/20 | 235.0 | 4.0 | 1803.987 |
| 95 | Asia | Afghanistan | 04/04/20 | 235.0 | 4.0 | 1803.987 |
| 96 | Asia | Afghanistan | 05/04/20 | 270.0 | 5.0 | 1803.987 |
| 97 | Asia | Afghanistan | 06/04/20 | 299.0 | 7.0 | 1803.987 |

```
In [ ]: #d. Fill all missing values with 0
```

```
In [198... var.fillna(0, inplace=True)  
var.head(111)
```

```
Out[198... 
```

|    | continent | location    | date     | total_cases | total_deaths | gdp_per_capita | h |
|----|-----------|-------------|----------|-------------|--------------|----------------|---|
| 0  | Asia      | Afghanistan | 31/12/19 | 0.0         | 0.0          | 1803.987       |   |
| 1  | Asia      | Afghanistan | 01/01/20 | 0.0         | 0.0          | 1803.987       |   |
| 2  | Asia      | Afghanistan | 02/01/20 | 0.0         | 0.0          | 1803.987       |   |
| 3  | Asia      | Afghanistan | 03/01/20 | 0.0         | 0.0          | 1803.987       |   |
| 4  | Asia      | Afghanistan | 04/01/20 | 0.0         | 0.0          | 1803.987       |   |
| 5  | Asia      | Afghanistan | 05/01/20 | 0.0         | 0.0          | 1803.987       |   |
| 6  | Asia      | Afghanistan | 06/01/20 | 0.0         | 0.0          | 1803.987       |   |
| 7  | Asia      | Afghanistan | 07/01/20 | 0.0         | 0.0          | 1803.987       |   |
| 8  | Asia      | Afghanistan | 08/01/20 | 0.0         | 0.0          | 1803.987       |   |
| 9  | Asia      | Afghanistan | 09/01/20 | 0.0         | 0.0          | 1803.987       |   |
| 10 | Asia      | Afghanistan | 10/01/20 | 0.0         | 0.0          | 1803.987       |   |
| 11 | Asia      | Afghanistan | 11/01/20 | 0.0         | 0.0          | 1803.987       |   |
| 12 | Asia      | Afghanistan | 12/01/20 | 0.0         | 0.0          | 1803.987       |   |
| 13 | Asia      | Afghanistan | 13/01/20 | 0.0         | 0.0          | 1803.987       |   |
| 14 | Asia      | Afghanistan | 14/01/20 | 0.0         | 0.0          | 1803.987       |   |
| 15 | Asia      | Afghanistan | 15/01/20 | 0.0         | 0.0          | 1803.987       |   |
| 16 | Asia      | Afghanistan | 16/01/20 | 0.0         | 0.0          | 1803.987       |   |
| 17 | Asia      | Afghanistan | 17/01/20 | 0.0         | 0.0          | 1803.987       |   |
| 18 | Asia      | Afghanistan | 18/01/20 | 0.0         | 0.0          | 1803.987       |   |
| 19 | Asia      | Afghanistan | 19/01/20 | 0.0         | 0.0          | 1803.987       |   |
| 20 | Asia      | Afghanistan | 20/01/20 | 0.0         | 0.0          | 1803.987       |   |
| 21 | Asia      | Afghanistan | 21/01/20 | 0.0         | 0.0          | 1803.987       |   |
| 22 | Asia      | Afghanistan | 22/01/20 | 0.0         | 0.0          | 1803.987       |   |
| 23 | Asia      | Afghanistan | 23/01/20 | 0.0         | 0.0          | 1803.987       |   |

|    |      |             |          |     |     |          |
|----|------|-------------|----------|-----|-----|----------|
| 26 | Asia | Afghanistan | 26/01/20 | 0.0 | 0.0 | 1803.987 |
| 27 | Asia | Afghanistan | 27/01/20 | 0.0 | 0.0 | 1803.987 |
| 28 | Asia | Afghanistan | 28/01/20 | 0.0 | 0.0 | 1803.987 |
| 29 | Asia | Afghanistan | 29/01/20 | 0.0 | 0.0 | 1803.987 |
| 30 | Asia | Afghanistan | 30/01/20 | 0.0 | 0.0 | 1803.987 |
| 31 | Asia | Afghanistan | 31/01/20 | 0.0 | 0.0 | 1803.987 |
| 32 | Asia | Afghanistan | 01/02/20 | 0.0 | 0.0 | 1803.987 |
| 33 | Asia | Afghanistan | 02/02/20 | 0.0 | 0.0 | 1803.987 |
| 34 | Asia | Afghanistan | 03/02/20 | 0.0 | 0.0 | 1803.987 |
| 35 | Asia | Afghanistan | 04/02/20 | 0.0 | 0.0 | 1803.987 |
| 36 | Asia | Afghanistan | 05/02/20 | 0.0 | 0.0 | 1803.987 |
| 37 | Asia | Afghanistan | 06/02/20 | 0.0 | 0.0 | 1803.987 |
| 38 | Asia | Afghanistan | 07/02/20 | 0.0 | 0.0 | 1803.987 |
| 39 | Asia | Afghanistan | 08/02/20 | 0.0 | 0.0 | 1803.987 |
| 40 | Asia | Afghanistan | 09/02/20 | 0.0 | 0.0 | 1803.987 |
| 41 | Asia | Afghanistan | 10/02/20 | 0.0 | 0.0 | 1803.987 |
| 42 | Asia | Afghanistan | 11/02/20 | 0.0 | 0.0 | 1803.987 |
| 43 | Asia | Afghanistan | 12/02/20 | 0.0 | 0.0 | 1803.987 |
| 44 | Asia | Afghanistan | 13/02/20 | 0.0 | 0.0 | 1803.987 |
| 45 | Asia | Afghanistan | 14/02/20 | 0.0 | 0.0 | 1803.987 |
| 46 | Asia | Afghanistan | 15/02/20 | 0.0 | 0.0 | 1803.987 |
| 47 | Asia | Afghanistan | 16/02/20 | 0.0 | 0.0 | 1803.987 |
| 48 | Asia | Afghanistan | 17/02/20 | 0.0 | 0.0 | 1803.987 |
| 49 | Asia | Afghanistan | 18/02/20 | 0.0 | 0.0 | 1803.987 |
| 50 | Asia | Afghanistan | 19/02/20 | 0.0 | 0.0 | 1803.987 |
| 51 | Asia | Afghanistan | 20/02/20 | 0.0 | 0.0 | 1803.987 |
| 52 | Asia | Afghanistan | 21/02/20 | 0.0 | 0.0 | 1803.987 |
| 53 | Asia | Afghanistan | 22/02/20 | 0.0 | 0.0 | 1803.987 |
| 54 | Asia | Afghanistan | 23/02/20 | 0.0 | 0.0 | 1803.987 |
| 55 | Asia | Afghanistan | 24/02/20 | 0.0 | 0.0 | 1803.987 |

|    |      |             |          |      |     |          |
|----|------|-------------|----------|------|-----|----------|
| 58 | Asia | Afghanistan | 27/02/20 | 1.0  | 0.0 | 1803.987 |
| 59 | Asia | Afghanistan | 28/02/20 | 1.0  | 0.0 | 1803.987 |
| 60 | Asia | Afghanistan | 29/02/20 | 1.0  | 0.0 | 1803.987 |
| 61 | Asia | Afghanistan | 01/03/20 | 1.0  | 0.0 | 1803.987 |
| 62 | Asia | Afghanistan | 02/03/20 | 1.0  | 0.0 | 1803.987 |
| 63 | Asia | Afghanistan | 03/03/20 | 0.0  | 0.0 | 1803.987 |
| 64 | Asia | Afghanistan | 04/03/20 | 0.0  | 0.0 | 1803.987 |
| 65 | Asia | Afghanistan | 05/03/20 | 0.0  | 0.0 | 1803.987 |
| 66 | Asia | Afghanistan | 06/03/20 | 0.0  | 0.0 | 1803.987 |
| 67 | Asia | Afghanistan | 07/03/20 | 0.0  | 0.0 | 1803.987 |
| 68 | Asia | Afghanistan | 08/03/20 | 4.0  | 0.0 | 1803.987 |
| 69 | Asia | Afghanistan | 09/03/20 | 0.0  | 0.0 | 1803.987 |
| 70 | Asia | Afghanistan | 10/03/20 | 0.0  | 0.0 | 1803.987 |
| 71 | Asia | Afghanistan | 11/03/20 | 7.0  | 0.0 | 1803.987 |
| 72 | Asia | Afghanistan | 12/03/20 | 0.0  | 0.0 | 1803.987 |
| 73 | Asia | Afghanistan | 13/03/20 | 0.0  | 0.0 | 1803.987 |
| 74 | Asia | Afghanistan | 14/03/20 | 0.0  | 0.0 | 1803.987 |
| 75 | Asia | Afghanistan | 15/03/20 | 10.0 | 0.0 | 1803.987 |
| 76 | Asia | Afghanistan | 16/03/20 | 16.0 | 0.0 | 1803.987 |
| 77 | Asia | Afghanistan | 17/03/20 | 21.0 | 0.0 | 1803.987 |
| 78 | Asia | Afghanistan | 18/03/20 | 22.0 | 0.0 | 1803.987 |
| 79 | Asia | Afghanistan | 19/03/20 | 22.0 | 0.0 | 1803.987 |
| 80 | Asia | Afghanistan | 20/03/20 | 22.0 | 0.0 | 1803.987 |
| 81 | Asia | Afghanistan | 21/03/20 | 24.0 | 0.0 | 1803.987 |
| 82 | Asia | Afghanistan | 22/03/20 | 24.0 | 0.0 | 1803.987 |
| 83 | Asia | Afghanistan | 23/03/20 | 34.0 | 0.0 | 1803.987 |
| 84 | Asia | Afghanistan | 24/03/20 | 40.0 | 1.0 | 1803.987 |
| 85 | Asia | Afghanistan | 25/03/20 | 42.0 | 1.0 | 1803.987 |
| 86 | Asia | Afghanistan | 26/03/20 | 75.0 | 1.0 | 1803.987 |

|     |      |             |          |       |      |          |
|-----|------|-------------|----------|-------|------|----------|
| 89  | Asia | Afghanistan | 29/03/20 | 106.0 | 3.0  | 1803.987 |
| 90  | Asia | Afghanistan | 30/03/20 | 114.0 | 4.0  | 1803.987 |
| 91  | Asia | Afghanistan | 31/03/20 | 141.0 | 4.0  | 1803.987 |
| 92  | Asia | Afghanistan | 01/04/20 | 166.0 | 4.0  | 1803.987 |
| 93  | Asia | Afghanistan | 02/04/20 | 192.0 | 4.0  | 1803.987 |
| 94  | Asia | Afghanistan | 03/04/20 | 235.0 | 4.0  | 1803.987 |
| 95  | Asia | Afghanistan | 04/04/20 | 235.0 | 4.0  | 1803.987 |
| 96  | Asia | Afghanistan | 05/04/20 | 270.0 | 5.0  | 1803.987 |
| 97  | Asia | Afghanistan | 06/04/20 | 299.0 | 7.0  | 1803.987 |
| 98  | Asia | Afghanistan | 07/04/20 | 337.0 | 7.0  | 1803.987 |
| 99  | Asia | Afghanistan | 08/04/20 | 367.0 | 11.0 | 1803.987 |
| 100 | Asia | Afghanistan | 09/04/20 | 423.0 | 14.0 | 1803.987 |
| 101 | Asia | Afghanistan | 10/04/20 | 484.0 | 15.0 | 1803.987 |
| 102 | Asia | Afghanistan | 11/04/20 | 521.0 | 15.0 | 1803.987 |
| 103 | Asia | Afghanistan | 12/04/20 | 555.0 | 18.0 | 1803.987 |
| 104 | Asia | Afghanistan | 13/04/20 | 607.0 | 18.0 | 1803.987 |
| 105 | Asia | Afghanistan | 14/04/20 | 665.0 | 21.0 | 1803.987 |
| 106 | Asia | Afghanistan | 15/04/20 | 714.0 | 23.0 | 1803.987 |
| 107 | Asia | Afghanistan | 16/04/20 | 784.0 | 25.0 | 1803.987 |
| 108 | Asia | Afghanistan | 17/04/20 | 794.0 | 29.0 | 1803.987 |
| 109 | Asia | Afghanistan | 18/04/20 | 845.0 | 30.0 | 1803.987 |
| 110 | Asia | Afghanistan | 19/04/20 | 908.0 | 30.0 | 1803.987 |

```
In [ ]: #6. Date time format :
        #a. Convert date column in datetime format using
        #pandas.to_datetime
```

```
In [208... var['date'] = pd.to_datetime(var['date'], format='%Y-%m-%d')
var.head()
```

|   |      |             |            |     |     |          |
|---|------|-------------|------------|-----|-----|----------|
| 0 | Asia | Afghanistan | 2019-12-31 | 0.0 | 0.0 | 1803.987 |
| 1 | Asia | Afghanistan | 2020-01-01 | 0.0 | 0.0 | 1803.987 |
| 2 | Asia | Afghanistan | 2020-02-01 | 0.0 | 0.0 | 1803.987 |
| 3 | Asia | Afghanistan | 2020-03-01 | 0.0 | 0.0 | 1803.987 |
| 4 | Asia | Afghanistan | 2020-04-01 | 0.0 | 0.0 | 1803.987 |

```
In [ ]: #b. Create new column month after extracting month data from
#date column.
```

```
In [212...] var['month'] = var['date'].dt.month
var.head()
```

```
Out[212...]
continent    location    date    total_cases    total_deaths    gdp_per_capita    humar
```

|   |      |             |            |     |     |          |
|---|------|-------------|------------|-----|-----|----------|
| 0 | Asia | Afghanistan | 2019-12-31 | 0.0 | 0.0 | 1803.987 |
| 1 | Asia | Afghanistan | 2020-01-01 | 0.0 | 0.0 | 1803.987 |
| 2 | Asia | Afghanistan | 2020-02-01 | 0.0 | 0.0 | 1803.987 |
| 3 | Asia | Afghanistan | 2020-03-01 | 0.0 | 0.0 | 1803.987 |
| 4 | Asia | Afghanistan | 2020-04-01 | 0.0 | 0.0 | 1803.987 |

```
In [214...] var['month'] = var['date'].dt.strftime('%B')
var.head()
```

|   |      |             |            |     |     |          |
|---|------|-------------|------------|-----|-----|----------|
| 0 | Asia | Afghanistan | 2019-12-31 | 0.0 | 0.0 | 1803.987 |
| 1 | Asia | Afghanistan | 2020-01-01 | 0.0 | 0.0 | 1803.987 |
| 2 | Asia | Afghanistan | 2020-02-01 | 0.0 | 0.0 | 1803.987 |
| 3 | Asia | Afghanistan | 2020-03-01 | 0.0 | 0.0 | 1803.987 |
| 4 | Asia | Afghanistan | 2020-04-01 | 0.0 | 0.0 | 1803.987 |

```
In [ ]: # 7. Data Aggregation:
# a. Find max value in all columns using groupby function on continent'col
# b. Store the result in a new dataframe named 'df_groupby'.(Use df_groupby)
```

```
In [ ]:
```

```
In [218...] max_values = var.groupby('continent').max().reset_index()
print(max_values)
```

|   | continent     | location                     | date       | total_cases | \ |
|---|---------------|------------------------------|------------|-------------|---|
| 0 | Africa        | Zimbabwe                     | 2020-12-11 | 752269.0    |   |
| 1 | Asia          | Yemen                        | 2020-12-11 | 8874290.0   |   |
| 2 | Europe        | Vatican                      | 2020-12-11 | 1991233.0   |   |
| 3 | North America | United States Virgin Islands | 2020-12-11 | 11205486.0  |   |
| 4 | Oceania       | Wallis and Futuna            | 2020-12-11 | 27750.0     |   |
| 5 | South America | Venezuela                    | 2020-12-11 | 5876464.0   |   |

|   | total_deaths | gdp_per_capita | human_development_index | month     |
|---|--------------|----------------|-------------------------|-----------|
| 0 | 20314.0      | 26382.287      | 0.797                   | September |
| 1 | 130519.0     | 116935.600     | 0.933                   | September |
| 2 | 52147.0      | 94277.965      | 0.953                   | September |
| 3 | 247220.0     | 54225.446      | 0.926                   | September |
| 4 | 907.0        | 44648.710      | 0.939                   | September |
| 5 | 166014.0     | 22767.037      | 0.843                   | September |

```
In [227...] df_groupby = var.groupby('continent').max().reset_index()
print(df_groupby)
```

|   |             |                              |            |            |
|---|-------------|------------------------------|------------|------------|
| 1 | Asia        | Yemen                        | 2020-12-11 | 8874290.0  |
| 2 | Europe      | Vatican                      | 2020-12-11 | 1991233.0  |
| 3 | North erica | United States Virgin Islands | 2020-12-11 | 11205486.0 |
| 4 | Oceania     | Wallis and Futuna            | 2020-12-11 | 27750.0    |
| 5 | South erica | Venezuela                    | 2020-12-11 | 5876464.0  |

|   | total_deaths | gdp_per_capita | human_develo | ent_index | month     |
|---|--------------|----------------|--------------|-----------|-----------|
| 0 | 20314.0      | 26382.287      |              | 0.797     | September |
| 1 | 130519.0     | 116935.600     |              | 0.933     | September |
| 2 | 52147.0      | 94277.965      |              | 0.953     | September |
| 3 | 247220.0     | 54225.446      |              | 0.926     | September |
| 4 | 907.0        | 44648.710      |              | 0.939     | September |
| 5 | 166014.0     | 22767.037      |              | 0.843     | September |

In [223... *#8. Feature Engineering :*

In [ ]: *#a. Create a new feature 'total\_deaths\_to\_total\_cases' by ratio of 'total\_*

```
In [233... var['total_cases'].replace(0, pd.NA, inplace=True)

var['total_deaths_to_total_cases'] = var['total_deaths'] / var['total_cas
print(var.head())
```

|   | continent | location    | date       | total_cases | total_deaths | gdp_per_cap |
|---|-----------|-------------|------------|-------------|--------------|-------------|
| 0 | Asia      | Afghanistan | 2019-12-31 | <NA>        | 0.0          | 1803.       |
| 1 | Asia      | Afghanistan | 2020-01-01 | <NA>        | 0.0          | 1803.       |
| 2 | Asia      | Afghanistan | 2020-02-01 | <NA>        | 0.0          | 1803.       |
| 3 | Asia      | Afghanistan | 2020-03-01 | <NA>        | 0.0          | 1803.       |
| 4 | Asia      | Afghanistan | 2020-04-01 | <NA>        | 0.0          | 1803.       |

|   | human_develo | ent_index | month    | total_deaths_to_total_cases |
|---|--------------|-----------|----------|-----------------------------|
| 0 |              | 0.498     | December | <NA>                        |
| 1 |              | 0.498     | January  | <NA>                        |
| 2 |              | 0.498     | February | <NA>                        |
| 3 |              | 0.498     | March    | <NA>                        |
| 4 |              | 0.498     | April    | <NA>                        |

In [235... var.tail()

|       |        |          |            |        |       |          |
|-------|--------|----------|------------|--------|-------|----------|
| 56743 | Africa | Zimbabwe | 2020-11-13 | 8696.0 | 255.0 | 1899.775 |
| 56744 | Africa | Zimbabwe | 2020-11-14 | 8765.0 | 257.0 | 1899.775 |
| 56745 | Africa | Zimbabwe | 2020-11-15 | 8786.0 | 257.0 | 1899.775 |
| 56746 | Africa | Zimbabwe | 2020-11-16 | 8786.0 | 257.0 | 1899.775 |
| 56747 | Africa | Zimbabwe | 2020-11-17 | 8897.0 | 257.0 | 1899.775 |

In [237... *#9. Data Visualization :*  
*#a. Perform Univariate analysis on 'gdp\_per\_capita' column by*  
*#plotting histogram using seaborn dist plot.*

```
import seaborn as sns
import matplotlib.pyplot as plt

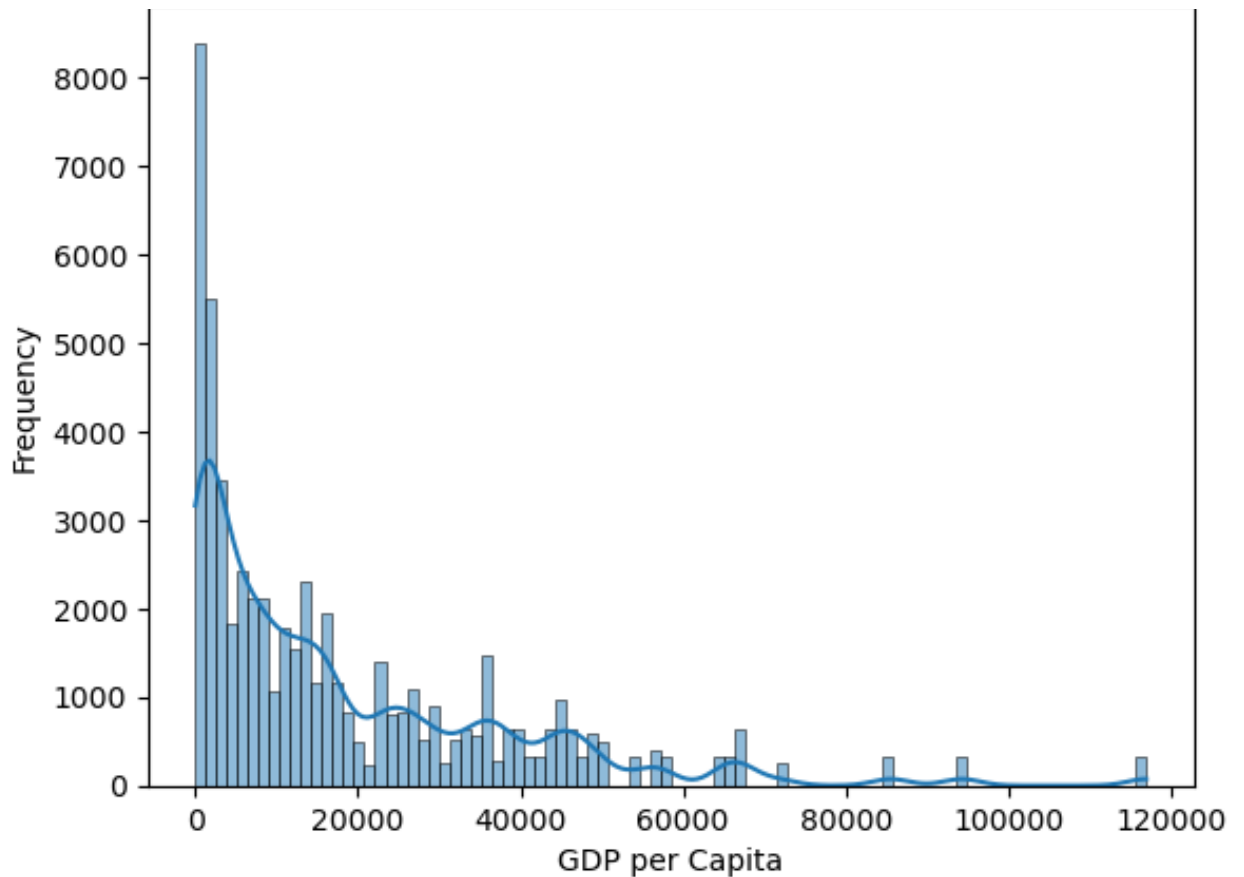
# Plotting the histogram with a kernel density estimate (KDE)
sns.histplot(var['gdp_per_capita'], kde=True)

# Adding titles and labels
plt.title('Distribution of GDP per Capita')
plt.xlabel('GDP per Capita')
plt.ylabel('Frequency')

# Displaying the plot
plt.show()
```

Matplotlib is building the font cache; this may take a moment.  
/opt/conda/envs/anaconda-2024.02-py310/lib/python3.10/site-packages/seaborn/\_oldcore.py:1119: FutureWarning: use\_inf\_as\_na option is deprecated and will be removed in a future version. Convert inf values to NaN before operating instead.

```
with pd.option_context('mode.use_inf_as_na', True):
```

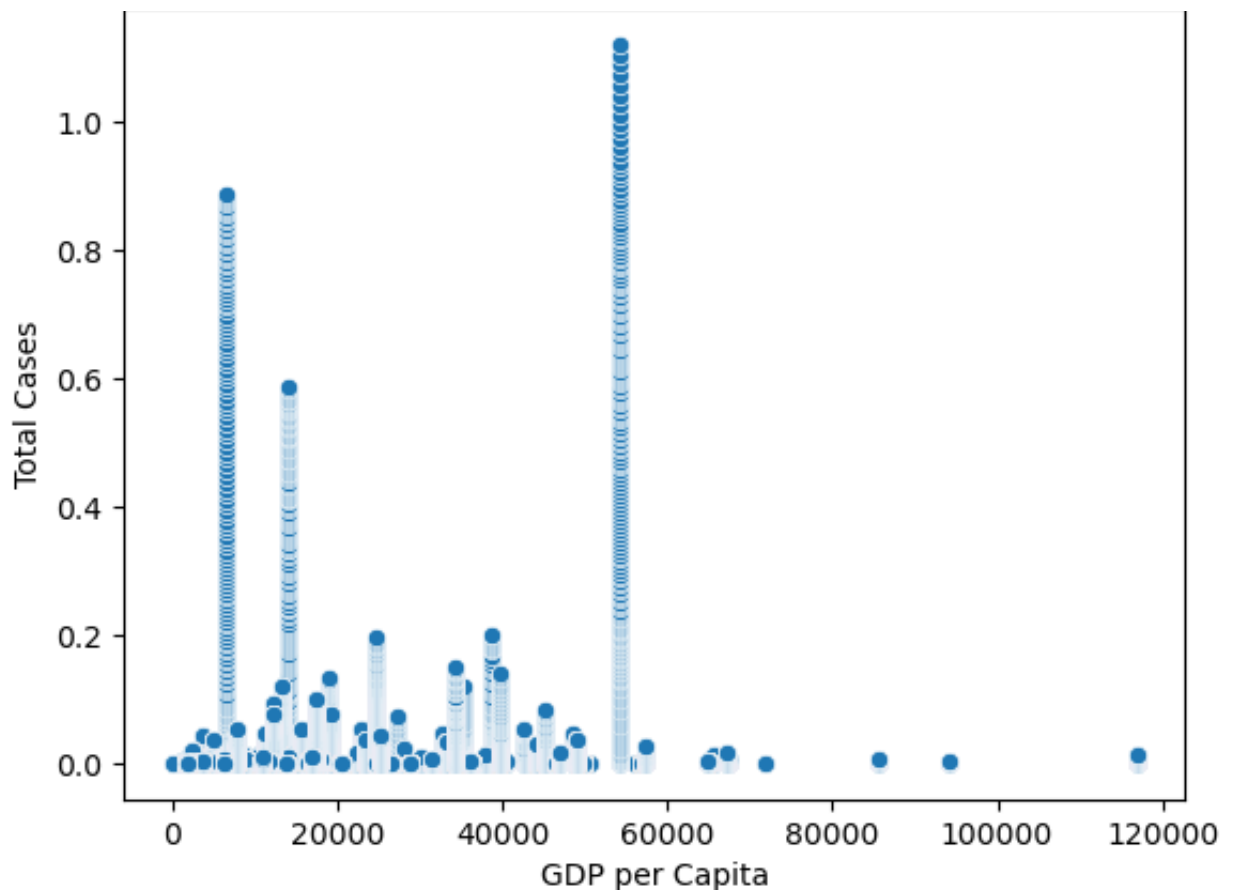


```
In [242...] #b. Plot a scatter plot of 'total_cases' & 'gdp_per_capita'
var['gdp_per_capita'].fillna(var['gdp_per_capita'].median(), inplace=True)
var['total_cases'].fillna(var['total_cases'].median(), inplace=True)
```

```
In [244...] sns.scatterplot(data=var, x='gdp_per_capita', y='total_cases')

plt.title('Total Cases vs. GDP per Capita')
plt.xlabel('GDP per Capita')
plt.ylabel('Total Cases')

plt.show()
```



In [283... *#c. Plot Pairplot on df\_groupby dataset.*

```
sns.pairplot(data=df_groupby, height=3.5 , aspect=1)

plt.show()
```

/opt/conda/envs/anaconda-2024.02-py310/lib/python3.10/site-packages/seaborn/\_oldcore.py:1119: FutureWarning: use\_inf\_as\_na option is deprecated and will be removed in a future version. Convert inf values to NaN before operating instead.

```
with pd.option_context('mode.use_inf_as_na', True):
```

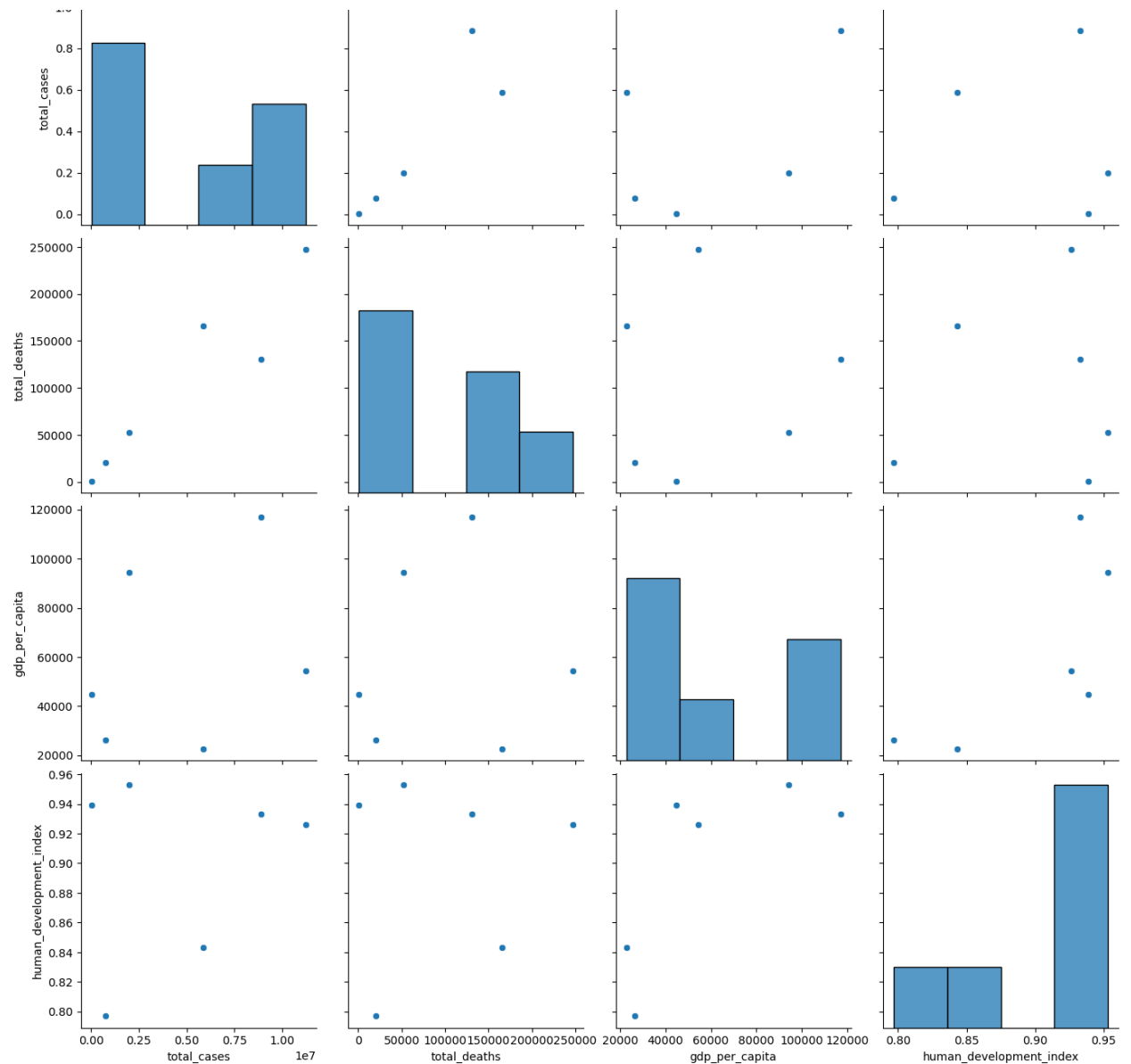
/opt/conda/envs/anaconda-2024.02-py310/lib/python3.10/site-packages/seaborn/\_oldcore.py:1119: FutureWarning: use\_inf\_as\_na option is deprecated and will be removed in a future version. Convert inf values to NaN before operating instead.

```
with pd.option_context('mode.use_inf_as_na', True):
```

/opt/conda/envs/anaconda-2024.02-py310/lib/python3.10/site-packages/seaborn/\_oldcore.py:1119: FutureWarning: use\_inf\_as\_na option is deprecated and will be removed in a future version. Convert inf values to NaN before operating instead.

```
with pd.option_context('mode.use_inf_as_na', True):
```

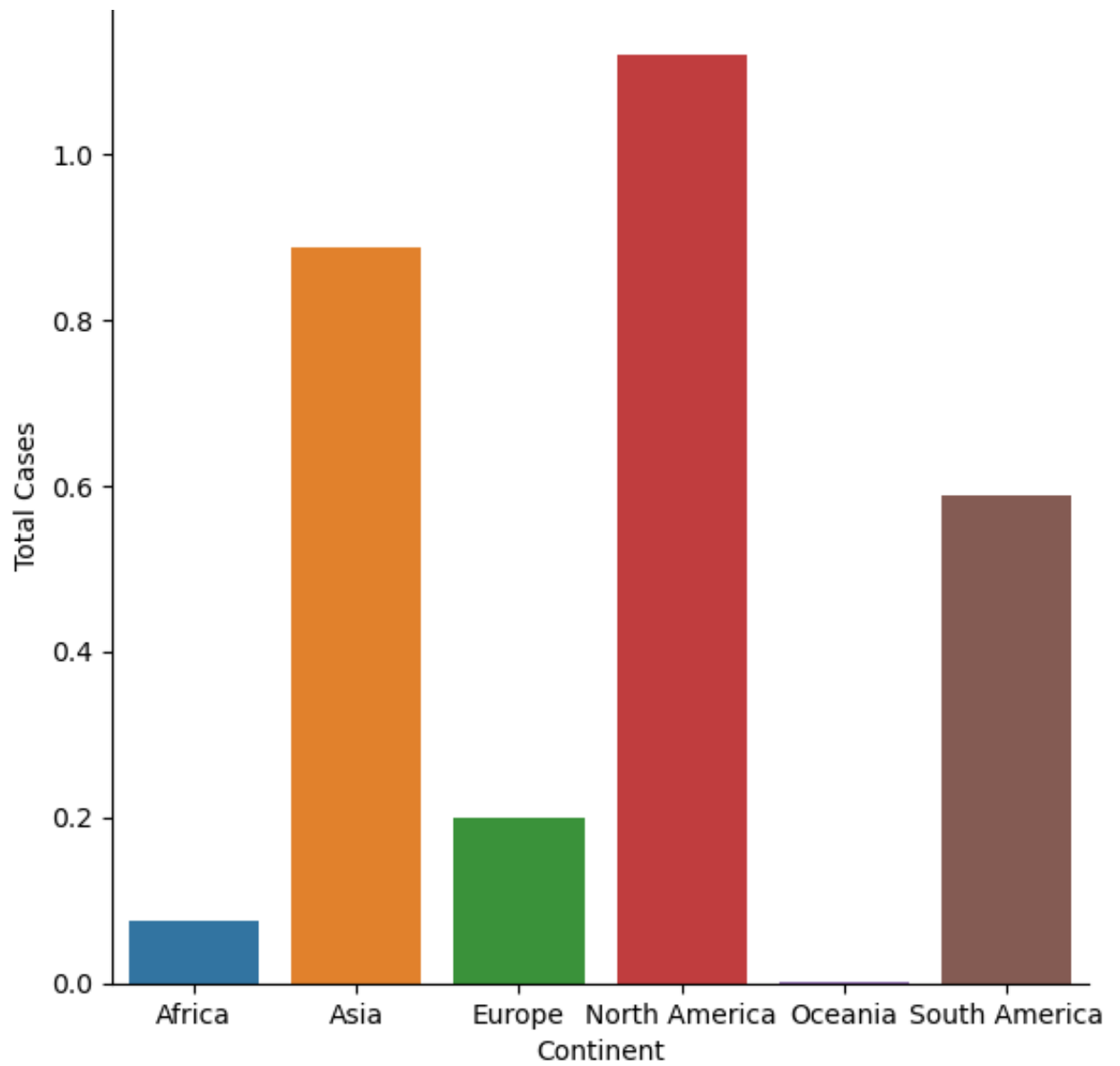
/opt/conda/envs/anaconda-2024.02-py310/lib/python3.10/site-packages/seaborn/\_oldcore.py:1119: FutureWarning: use\_inf\_as\_na option is deprecated and will be removed in a future version. Convert inf values to NaN before operating instead.



```
In [267... # Plot bar plot using catplot
sns.catplot(data=df_groupby, x='continent', y='total_cases', kind='bar',

# Add labels and title
plt.title("Total COVID-19 Cases per Continent")
plt.xlabel("Continent")
plt.ylabel("Total Cases")

plt.show()
```



In [ ]:

In [ ]: